
IPL Analysis

Toss impact, phases, and player performance
using Python



Introduction

Analyze IPL ball-by-ball data to assess *toss impact*, quantify contributions of *powerplay*, *middle*, and *death overs*, and identify top batters and bowlers.





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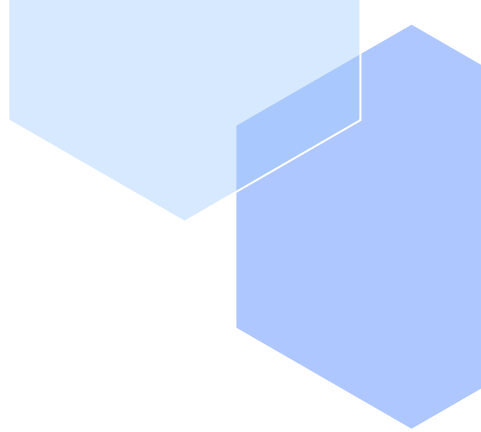
Toss Analysis

Objectives: toss impact

Determine whether winning the toss affects match outcomes and quantify effect size across seasons.

Compare win percentages for *toss winners* vs *toss losers* and highlight practical significance; key takeaway: **negligible advantage**.

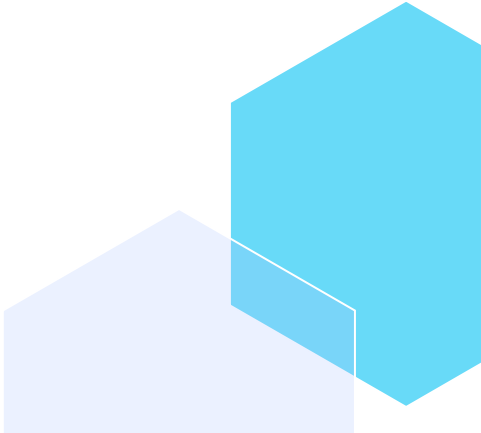
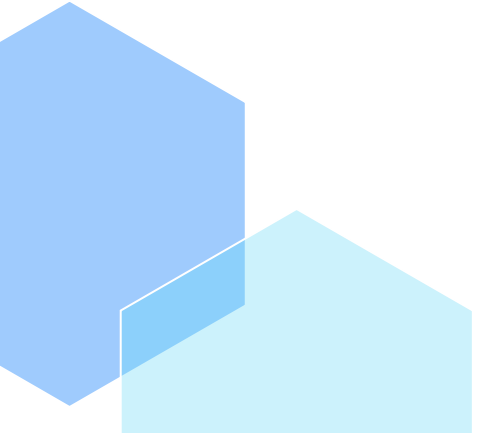


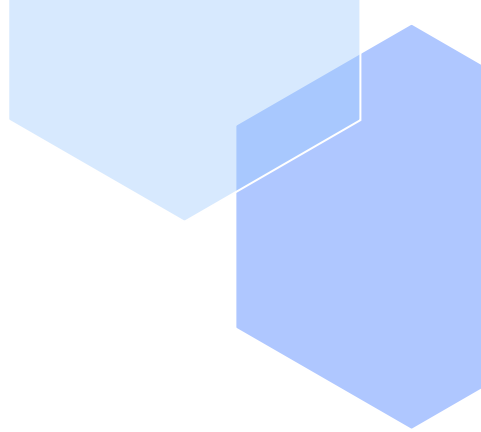


Methodology: ball-by-ball, Python

Use ball-by-ball IPL dataset aggregated by match and phase. Clean and feature-engineer with Pandas, compute phase scores, and generate charts with Matplotlib/seaborn.

Produce bar and grouped-bar charts, leaderboards for runs/wickets, and concise dashboard visuals for executive review.

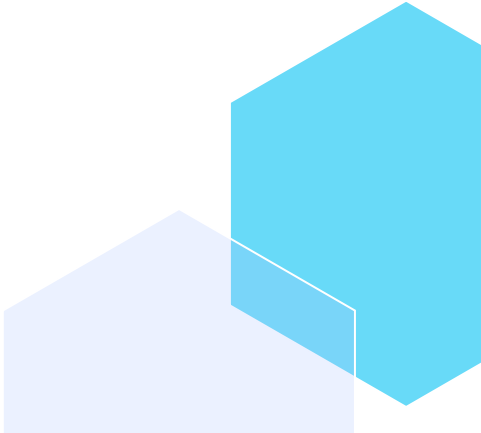
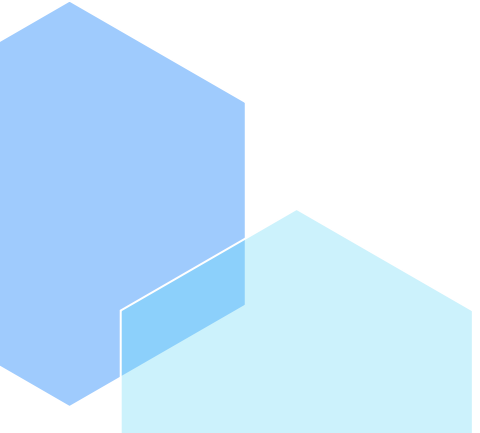




Key insight: negligible advantage

Winning the toss yields only a slight edge — toss winners win **50.4%** of matches, effectively a coin flip.

The *middle overs* show the largest scoring gap between winners and losers, making phase performance more decisive than the toss.



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Player Performance

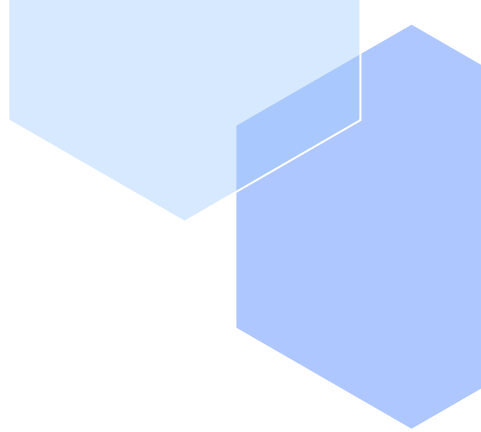


Top batters: runs leaderboard

Ranked by aggregate runs across seasons; report highlights *Top 5* batters with runs, strike rate, and consistency score.

Use concise tables and sparkline trends to show each batter's seasonal peaks and contribution to wins.

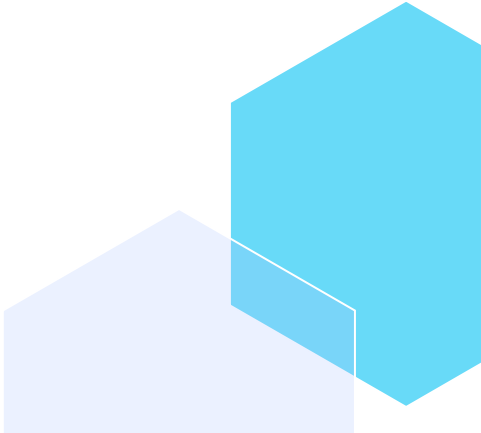
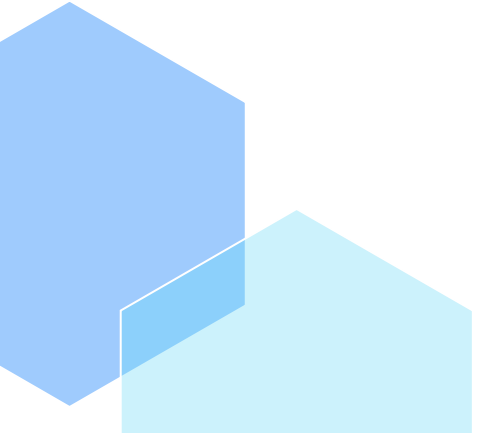




Top bowlers: wickets leaderboard

Ranked by total wickets and wickets-per-match; present *Top 5* bowlers with economy and strike-rate metrics.

Include role-based breakdown (death overs vs middle overs) to surface situational impact.

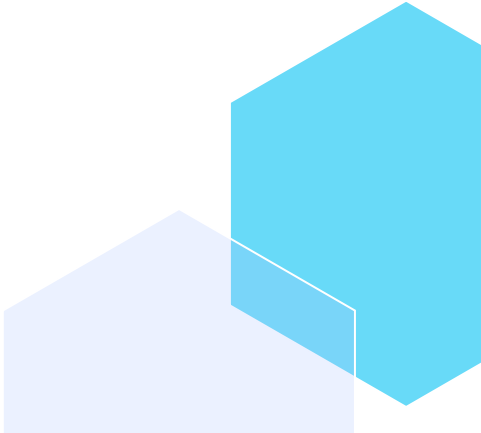
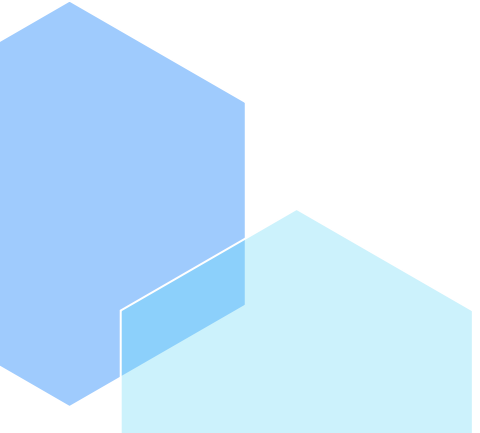


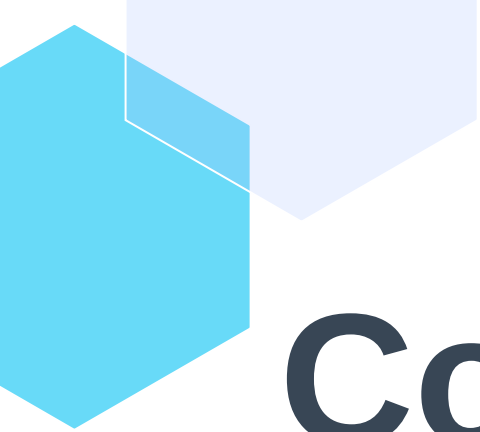


Season trends: consistency & peaks

Analyze seasonal trajectories to identify sustained performers and breakout years using rolling averages.

Visualize with heatmaps and line charts to reveal consistency, peaks, and form volatility.





Conclusions

Toss outcome has **minimal influence**; teams should prioritize phase performance, especially the *middle overs*.

Recommended actions: focus coaching and strategy on middle-over control, and use identified top performers for match planning. Tools: **Python, Pandas, Matplotlib** for reproducible dashboards.

